

CS & IT ENGINEERING



COMPUTER ORGANIZATION AND ARCHITECTURE

Instruction & Addressing Modes

Lecture No.- 01

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Recap of Previous Lecture



Topic

Micro Operation

Topic

Memory Access

Topics to be Covered



Topic

Instruction

$$\text{floor}(13.6) = 13$$

$$\text{floor}(2.9) = 2$$

$$\text{floor}(3.1) = 3$$

Topic

ISA

$$\text{ceil}(13.6) = 14$$

$$\text{ceil}(2.9) = 3$$

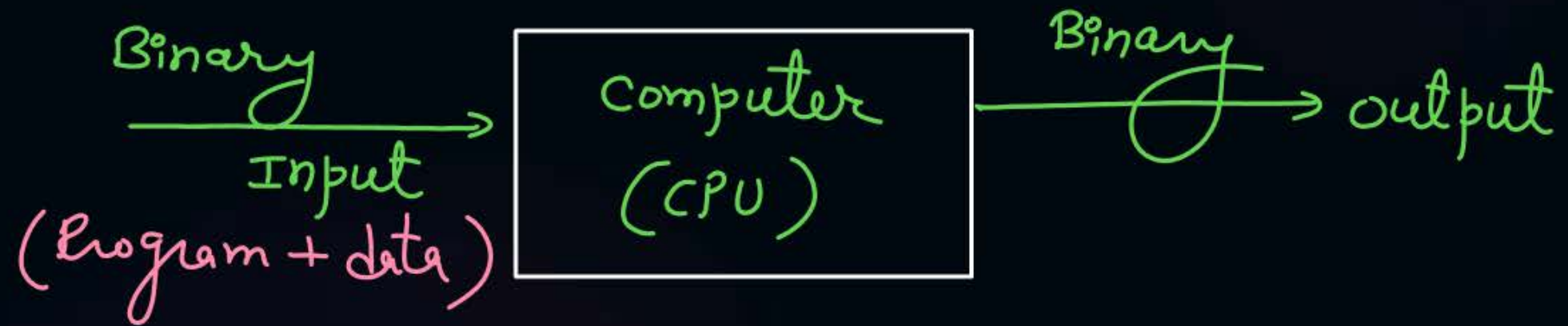
$$\text{ceil}(3.1) = 4$$

Topic

Types of Instruction



Topic : Digital Computer





Topic : Instruction

High level language code

```
#include<stdio.h>
void main()
{
    int a, b, c;
    printf("Enter 2 values: ");
    scanf("%d %d", &a, &b);
    c = a + b;
    printf("Sum = %d", c);
}
```

Programming
statement

Compiler

Language Translation

Low level language code
or
Binary code

Instruction

1 0 1 1 1 0 0 0

1 0 0 0 0 0 0 1

1 1 1 1 0 0 1 0

0 1 0 1 0 1 0 1

1 1 1 1 0 1 1 0

0 1 0 1 0 1 0 1

1 0 0 0 1 1 1 1

1 0 1 0 0 0 1 1

0 0 1 1 1 1 0 1

or
machine code
or
byte code



Topic : Instruction

A group of bits which instructs computer to perform some operation

operation code	operand information
-------------------	---------------------

or
opcode

assume,
8 distinct operations $\Rightarrow$

	opcode	
ADD	000	3 bits in opcode
SUB	001	
INC	010	
DEC	011	
MOV	100	
Comp.	101	
MUL	110	
DIV	111	



Topic : Instruction

no. of distinct operations
or
no. of distinct type instructions
supported by CPU

opcode

8

3 bits

16

4 bits

13

4 bits

n

$\lceil \log_2 n \rceil$ bits



Topic : ISA



(instruction set architecture)

The set or collection of all instructions a CPU can support.

↓
type and format

size of ISA
or
size of instⁿ set

} no. of distinct instructions supported by CPU.



Topic : Types of Instruction



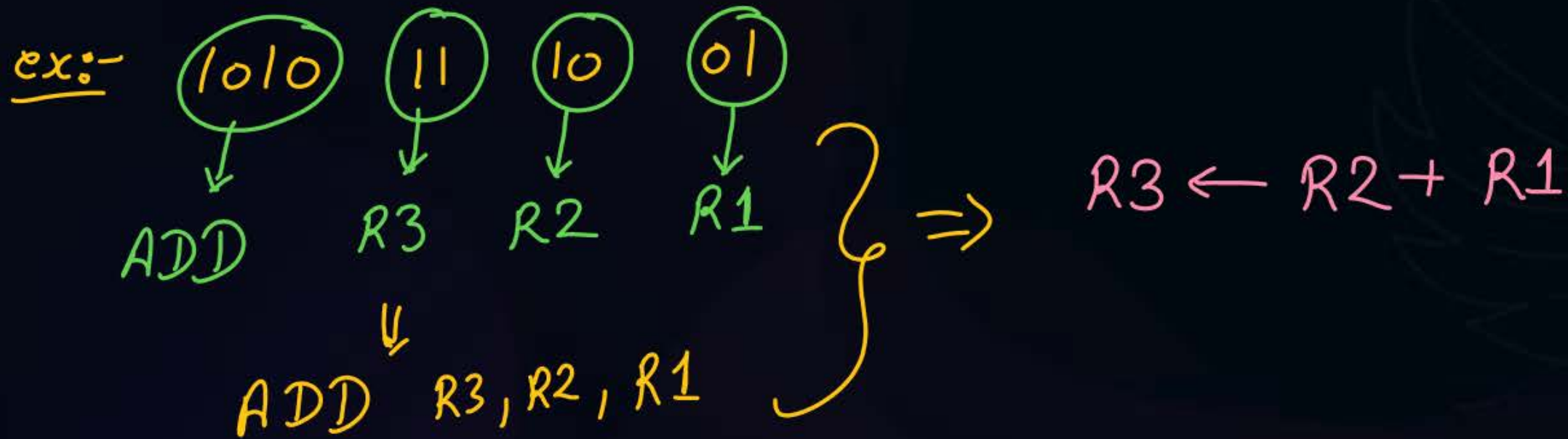
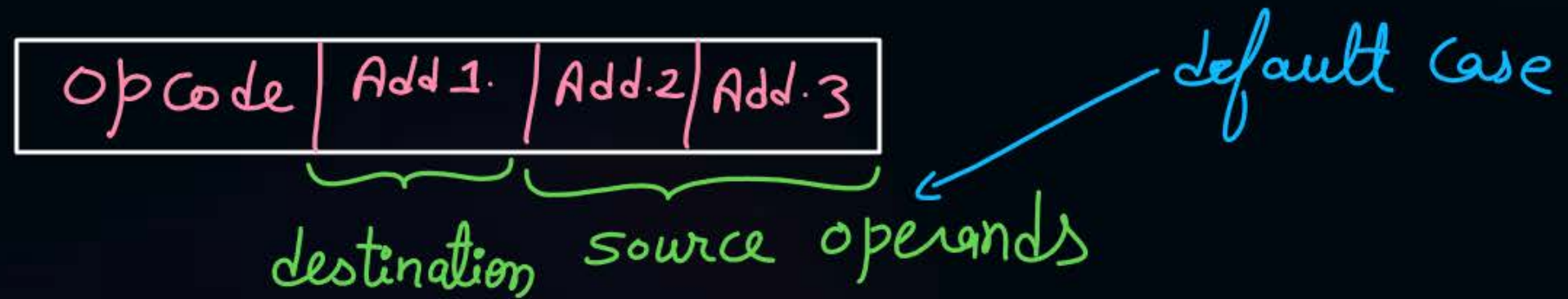
- 3-Address Instruction:
- 2-Address Instruction:
- 1-Address Instruction:
- 0-Address Instruction:





Topic : 3-Address Instruction

Max 3 addresses can be specified within an instruction *for operands*



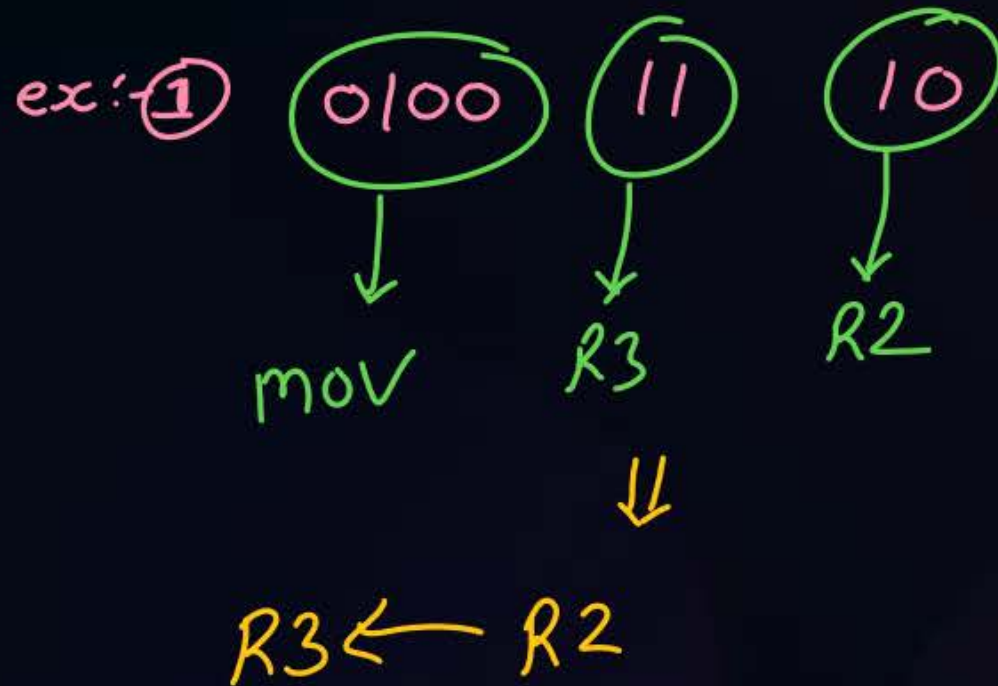


Topic : 2-Address Instruction

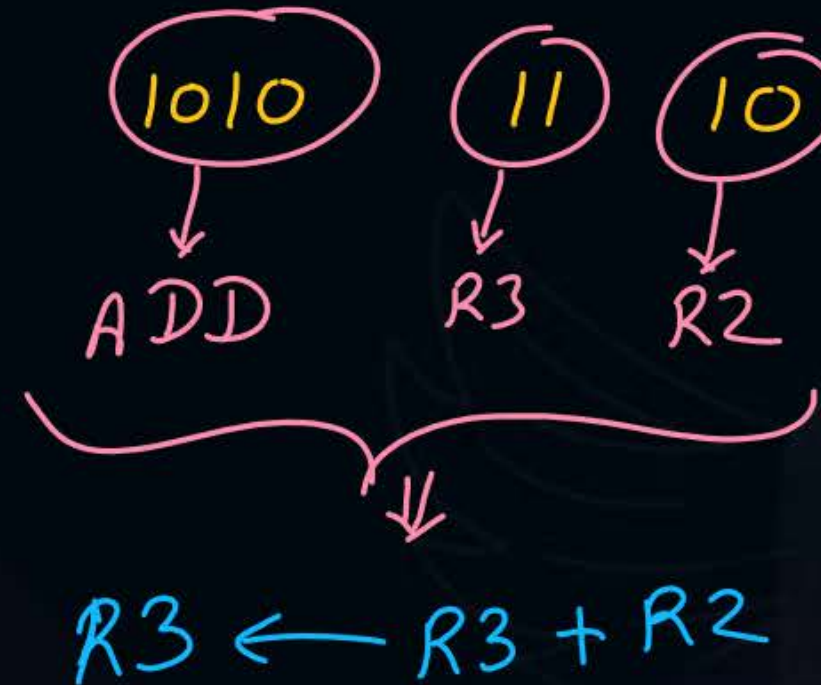
Max 2 addresses can be specified within an instruction

destination

opcode	Add.1	Add.2
--------	-------	-------



ex:- 2

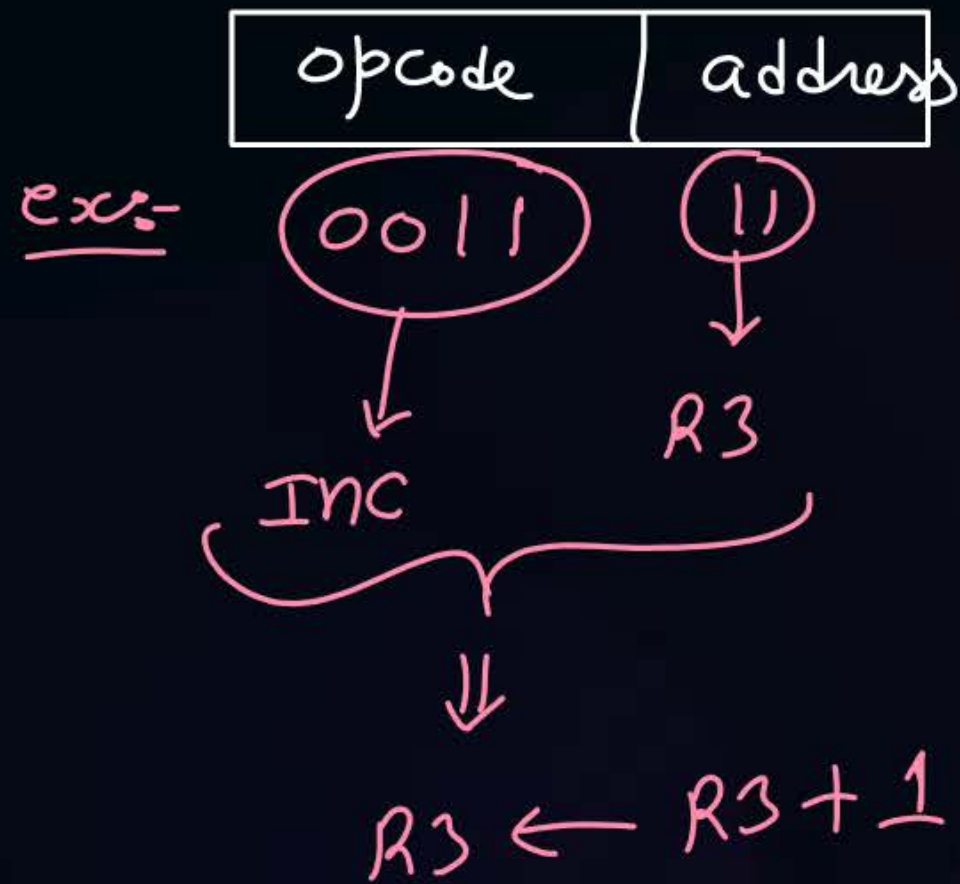


one of the 2 operands is used as source and destination both.



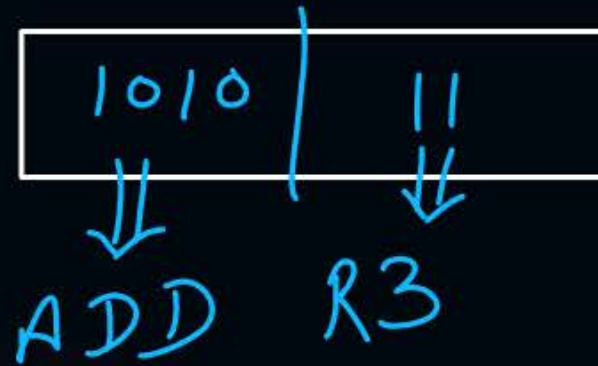
Topic : 1-Address Instruction

Max 1 address can be specified within an instruction

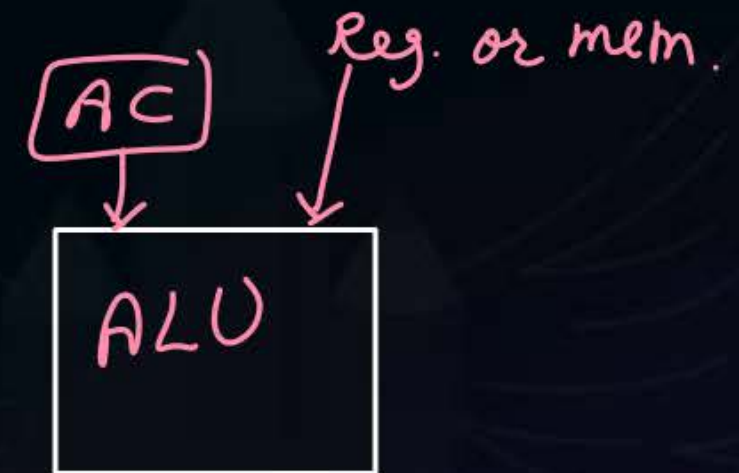


AC-based architecture

↓
it supports only max 1-add-type instns



$$AC \leftarrow AC + R3$$





Topic : 0-Address Instruction



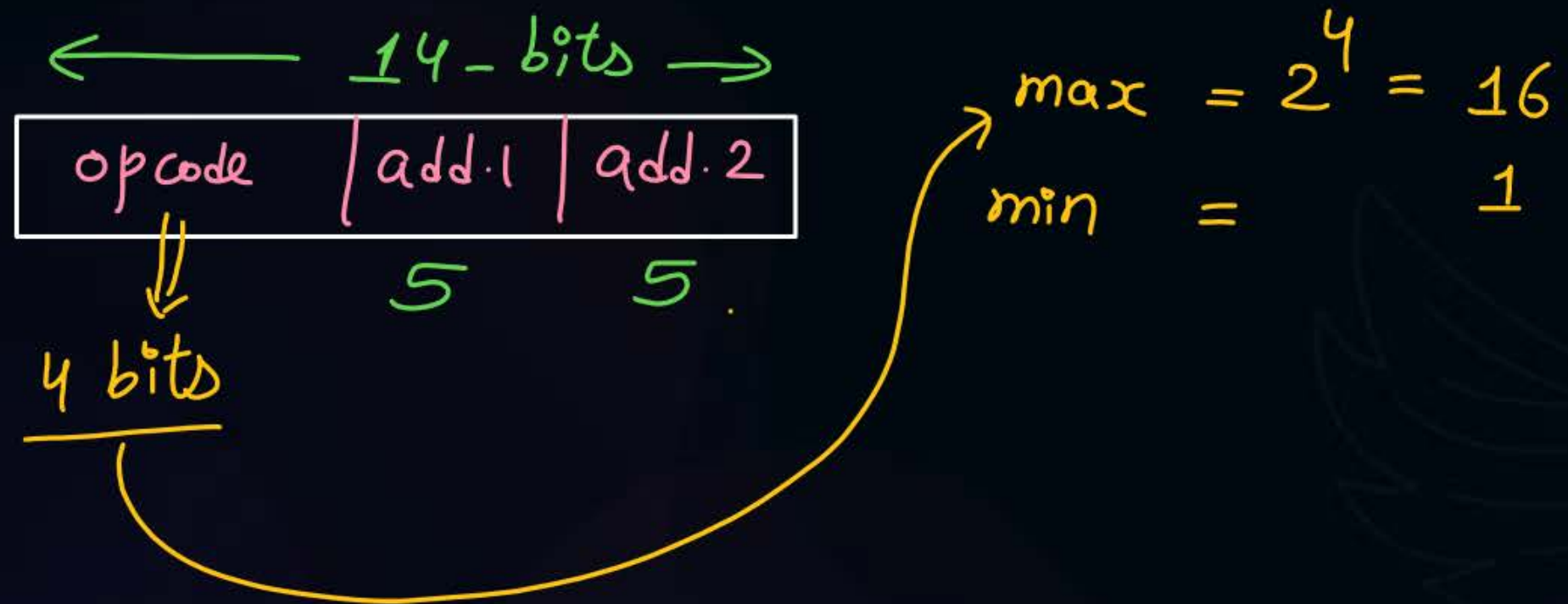
No any address can be specified within an instruction

opcode

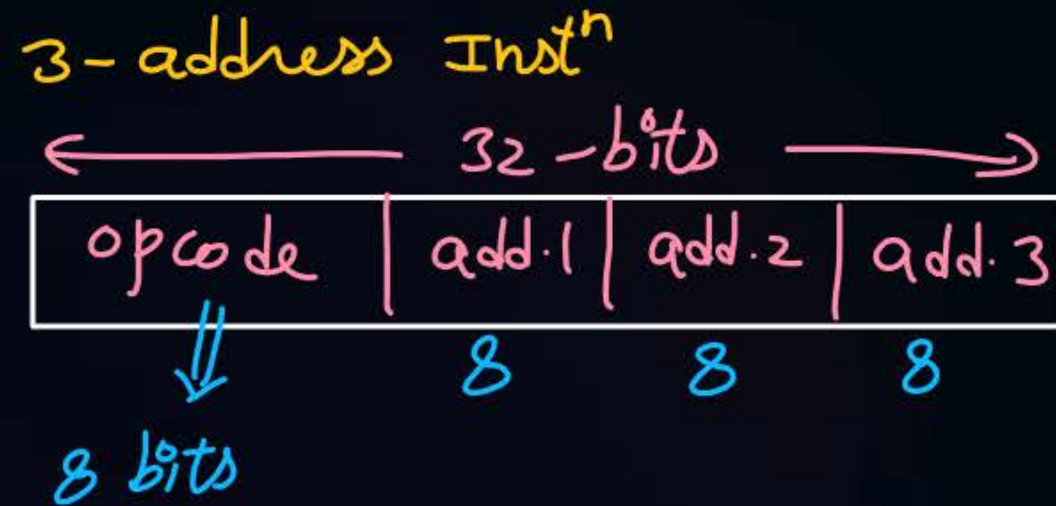
ex:- HALT



#Q. Consider a digital computer which supports only 2-address instructions each with 14-bits. If address length is 5-bits then maximum and minimum how many instructions the system can support?



#Q. Consider a digital computer which supports only 3-address instructions each with 32-bits. If address length is 8-bits then maximum and minimum how many instructions the system can support?



$$\begin{aligned} \text{max} &= 2^8 = 256 \\ \text{min} &= 1 \end{aligned}$$

#Q. Consider a digital computer which supports 64 2-address instructions. If address length is 9-bits then the length of the instruction is 24 bits?

2-add. instⁿ:-

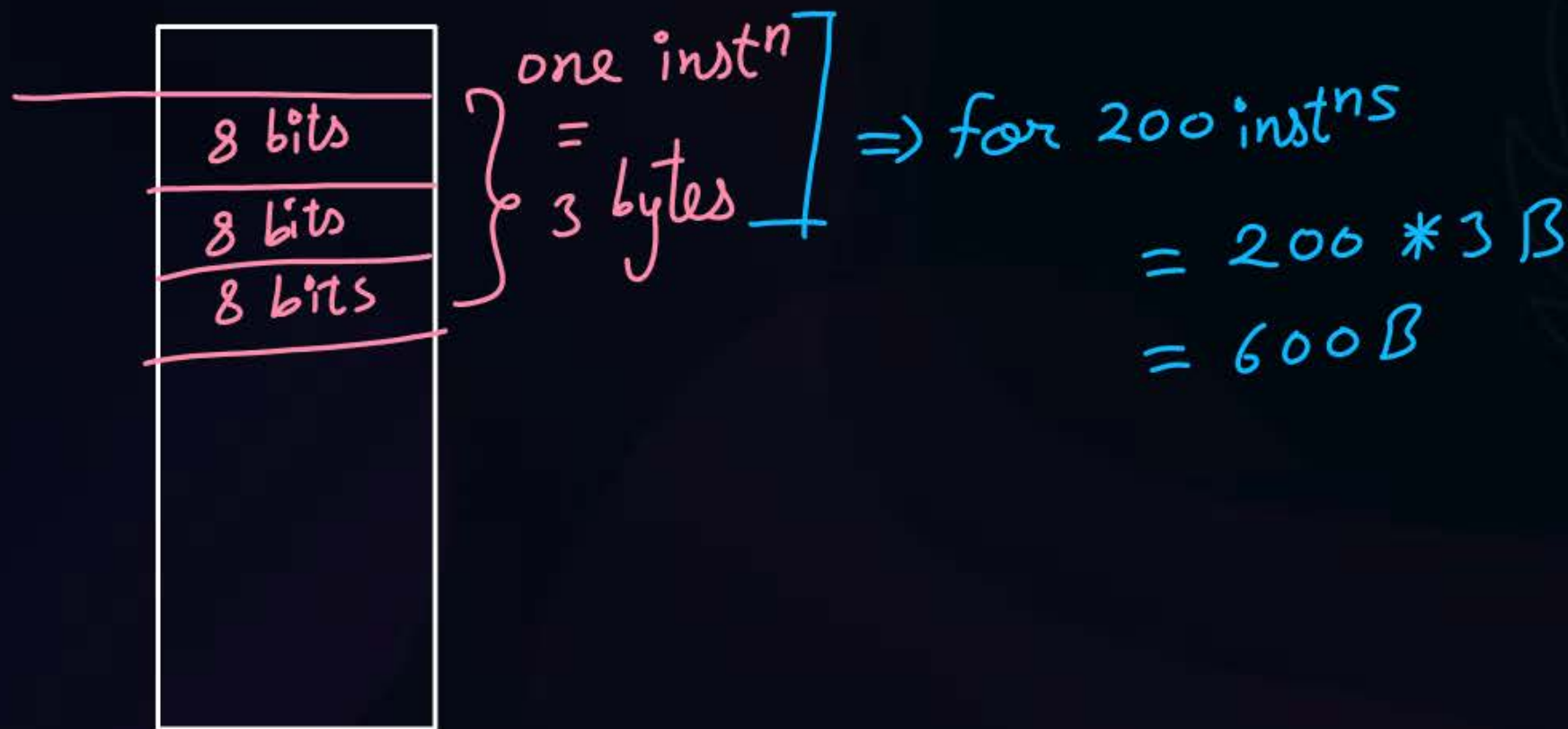


no. of instⁿs supported = 64

$$\downarrow$$
$$\text{opcode bits} = \lceil \log_2 64 \rceil = 6 \text{ bits}$$

#Q. Consider a digital computer which supports 64 2-address instructions. If address length is 9-bits then the length of the instruction is 24 bits?

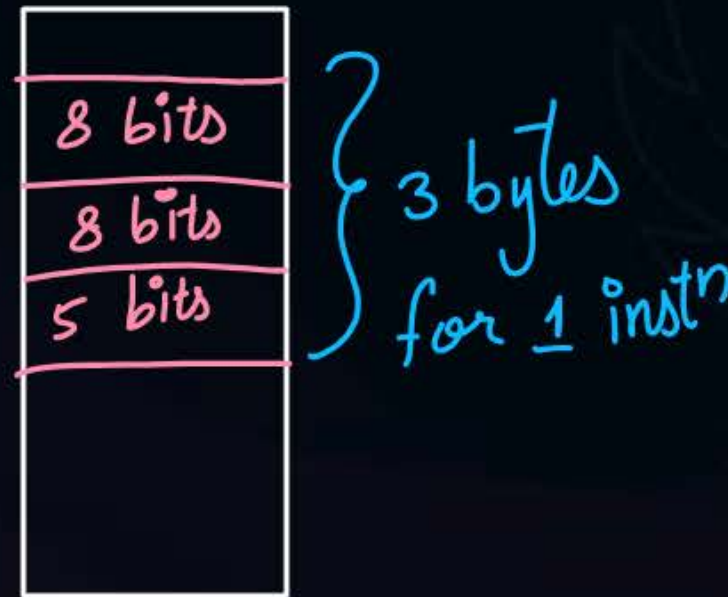
In above question: Each instruction must be stored in memory in a byte-aligned fashion. If a program has 200 instructions, then amount of memory required to store the program text is 600 bytes?



- #Q. Consider a digital computer which supports 32 2-address instructions. Consider the address length is 8-bits. Each instruction must be stored in memory in a byte-aligned fashion. If a program has 300 instructions, then amount of memory required to store the program text is 900 bytes?



no. of inst^{ns} = 32 $\Rightarrow$ opcode = 5 bits



then for 300 inst^{ns}
 $= 300 * 3B$
 $= 900B$

[NAT]

GATE-PYQ

H.W.



#Q. Consider the following instruction sequence where registers R1, R2 and R3 are general purpose and MEMORY [X] denotes the content at the memory location X.

Instruction	Semantics	Instruction Size (bytes)
MOV R1, (5000)	$R1 \leftarrow \text{MEMORY}[5000]$	4
MOV R2, (R3)	$R2 \leftarrow \text{MEMORY}[R3]$	4
ADD R2, R1	$R2 \leftarrow R1 + R2$	2
MOV (R3), R2	$\text{MEMORY}[R3] \leftarrow R2$	4
INC R3	$R3 \leftarrow R3 + 1$	2
DEC R1	$R1 \leftarrow R1 - 1$	2
BNZ 1004	Branch if not zero to the given absolute address	2
HALT	Stop	1

Assume that the content of the memory location 5000 is 10, and the content of the register R3 is 3000. The content of each of the memory locations from 3000 to 3010 is 50. The instruction sequence starts from the memory location 1000. All the numbers are in decimal format. Assume that the memory is byte addressable.

After the execution of the program, the content of memory location 3010 is 50.

$$R3 = 3000$$

1800	Instns
3000	50 60
3001	50 59
3002	50 58
	⋮
3009	50 51
3010	50
5000	10



2 mins Summary



Topic

Instructions

Topic

Instruction Set Architecture

Topic

Types of Instructions

Topic

Opcode



Happy Learning

THANK - YOU

